Samuel W Muwonge

AI Solutions Architect

AI Platforms & MLOps | Data Engineering & Automation | Generative AI & LLM Systems | Cloud-Native Solutions (Azure | AWS) | Finance & FinTech | Public Sector | Aerospace & Defence | Digital Transformation Enablement |

AI Solutions Architect helping organisations use artificial intelligence to work smarter, faster, and more securely. I design and deliver AI-powered systems that automate processes, reduce risk, and turn data into meaningful insights for better decisions. I work closely with business leaders and technical teams to ensure every solution supports real goals, improving efficiency, compliance, and customer experience.

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Nationality

BRITISH

Core Data Tech Stack

- **Storage & Lakehouse:** Azure Data Lake Storage Gen2 (ADLS), Delta Lake for ACID transactions, Azure Synapse Analytics (Dedicated & Serverless SQL Pools), Azure SQL Database.
- **Ingestion & Orchestration:** Azure Data Factory (ADF) and Synapse Pipelines for data movement and orchestration, Azure Event Grid and Logic Apps for event-driven integration.
- **Processing / Transformations:** Azure Databricks (Apache Spark) for scalable ETL/ELT, notebooks, and Delta, using PySpark or Scala for Data processing, plus Azure Functions or Logic Apps, for lightweight transformations and automation.
- **Streaming & Real-Time Data:** Azure Event Hubs, Azure IoT Hub, and Azure Stream Analytics for ingestion and real-time event processing, with Databricks Structured Streaming for advanced streaming pipelines.
- **Serving & BI:** Synapse SQL serverless/Dedicated or Azure SQL Database as semantic/serving layers, Power BI for reports/dashboards (usually owned by analysts, but enable it)
- **Security, Governance, and Monitoring:** Microsoft Entra ID (Azure AD) for auth/ RBAC, Azure Key Vault for secrets, Microsoft Purview for catalog/lineage/ governance, Azure Monitor + Log Analytics for health, logs, and metrics
- **DevOps & Collaboration** Azure DevOps and GitHub for CI/CD of data pipelines and notebooks, Terraform and Bicep for Infrastructure as Code, Jira or Azure Boards for Agile delivery and sprint planning.
- **Collaboration & Project Management:** Azure Boards, Jira, Confluence, Azure DevOps, Slack, Microsoft Teams, Draw.io

Experience

OCT 2024 - Present
AI Solutions Architect | SymphonyAI

SymphonyAI delivers predictive and generative AI SaaS solutions across financial services, retail, and CPG. I lead the architecture and operationalisation of enterprise AI systems that combine LLMs, data pipelines, and automation frameworks to drive insight, compliance, and cost efficiency at scale.

Key Contributions at SymphonyAI (Financial Services vertical):

- Leading the architecture of the "Athena AI Platform," unifying LLM-based fund analysis, document automation, and compliance intelligence across private equity operations.
- Designing and implementing advanced RAG and vector retrieval architectures to deliver sub-second performance across multi-million-document datasets.
- Architecting agentic AI systems under "Project Helios" using LangGraph and LangMem, enabling autonomous multi-agent workflows for risk modelling and portfolio analytics.
- Integrating Azure OpenAI GPT-5 and Small Language Models (SLMs) into financial operations, reducing manual review time by 40% and optimising inference costs by 30%.
- Establishing AI governance and compliance frameworks ensuring transparency, explainability, and adherence to FCA and SEC standards.
- Mentoring cross-functional AI teams and driving best practices in prompt engineering, evaluation, and production-grade MLOps across distributed Azure environments.

Business Impact:

Driving SymphonyAI's transformation into a trusted AI partner for financial institutions by aligning AI architecture with strategic business outcomes. Collaborating directly with the CTO, Chief Data Officer, and senior product leadership to shape enterprise AI strategy, accelerate go-to-market delivery, and strengthen compliance assurance. Partnering with tier-one private equity firms, fund administrators, and regulatory technology providers to deliver scalable, secure, and cost-efficient AI solutions that enhance client confidence, operational transparency, and competitive differentiation.

JLY 2023 - OCT 2024
AI & Cloud Solutions Architect | Airbus Defence And Space

At Airbus Defence and Space, I designed sovereign AI architectures for classified workloads supporting defence missions and satellite operations. My focus was on secure cloud integration, data sovereignty, and AI observability across multi-domain networks.

Key Contributions to Airbus Defence and Space:

- Designed mission-ready AI telemetry pipelines in Azure Sovereign Cloud, ingesting and analysing satellite data for predictive maintenance and operational readiness.
- Built secure inference environments that allowed mission teams to run AI analytics on classified datasets while maintaining NATO and MoD sovereignty compliance.
- Integrated observability and drift detection systems using Falco, OpenTelemetry, and Grafana, enabling real-time detection of performance degradation in mission control systems.
- Automated deployment of defence-grade DevSecOps pipelines with Terraform, Bicep, and AKS, embedding MoD accreditation and cyber compliance controls.
- Collaborated with defence data scientists and intelligence teams to operationalise AI threat models for surveillance and situational awareness.
- Produced TOGAF and Archimate architecture models mapping AI and data workflows across allied systems, supporting secure collaboration with NATO partners.

Impact:

Delivered sovereign AI systems that enhanced mission resilience and interoperability, allowing Airbus to integrate AI into classified command environments without compromising national security.

Certifications

- Azure Data Engineer Associate
- Azure Administrator Associate
- CKA (Certified Kubernete Administrator)
- Docker Certified Associate
- HashiCorp Terraform Associate

Enterprise Trained

- Azure Security Engineer
- Databricks Data Engineer Associate
- Snowflake SnowPro Core
- Azure AI Fundamentals

Agile & Scrum Ways of Working

- **Participating in daily stand-ups**
  participating in daily stand-ups, sharing progress on pipelines and data transformations, highlighting blockers such as schema changes or failed ETL runs, and aligning with analysts, data scientists, and product teams on upcoming priorities.
- **Collaborating with stakeholders**
  product owners, business analysts, compliance, and operations teams, refining data requirements, ensuring data platforms are supporting regulatory and business needs, and translating those requirements into technical delivery.
- **Contributing to sprint planning**
  estimating and prioritising data engineering tasks such as onboarding new sources, optimising pipelines, or building curated datasets, making sure work is aligning with development and reporting objectives.
- **Showcasing work during sprint reviews/demo days**
  e.g., demonstrating new pipelines, curated datasets, or streaming solutions, highlighting performance improvements, automated validations, or real-time insights that are adding business value.
- **Retrospectives participation**
  providing feedback on tooling, pipeline reliability, and delivery bottlenecks, suggesting improvements such as enhanced monitoring, CI/CD adoption, or better schema handling to improve future sprints.

APR 2022 - JUL 2023

AI Platform Engineer | DWP Digital

DWP Digital underpins the UK's largest citizen-facing platforms, including Universal Credit, Pensions, and Benefits. I helped design AI and automation solutions to improve fraud detection, service delivery, and transparency within welfare systems.

Key Contributions to DWP Digital Services:

- Developed and deployed real-time AI pipelines using Kafka and Spark, enabling proactive fraud detection across welfare and pension claim systems.
- Built AI model orchestration frameworks on Azure Machine Learning and MLflow, automating training, deployment, and retraining across multiple citizen-facing services.
- Created explainability dashboards integrating Grafana and OpenAI APIs, helping policy leaders visualise and justify AI-driven decisions in compliance with GDS standards.
- Automated infrastructure provisioning with Terraform and GitHub Actions, improving release reliability for AI models without service downtime.
- Worked cross-functionally with policy, risk, and technology teams to align data ethics and algorithmic fairness with DWP's public-service mission.
- Documented and modelled AI ecosystems in Archimate, mapping interactions between data, services, and policy to support long-term AI strategy.

Impact: Improved accuracy and fairness of welfare processing, reduced fraud losses, and enhanced public trust in DWP's use of AI for citizen services.

MAY 2020 - APR 2022

AI Systems Engineer | Monzo Bank

At Monzo, one of the UK's leading digital banks, I contributed to the development of AI systems that improved fraud detection, compliance automation, and customer experience through data-driven intelligence.

Key Contributions to Monzo Digital Bank:

- Engineered real-time fraud detection pipelines using Kafka, Flink, and PyTorch, processing millions of transactions daily to identify anomalies instantly.
- Implemented NLP-powered support automation using Hugging Face Transformers, cutting manual query resolution time for operations teams.
- Built explainability tooling and model-monitoring dashboards for FCA audits, providing visibility into model logic and bias detection.
- Integrated MLOps practices through GitHub Actions and MLflow, ensuring reproducibility and traceability for all production models.
- Collaborated with risk and compliance teams to align AI pipeline design with financial regulatory frameworks.
- Contributed to data-driven product innovation, enabling Monzo to roll out new personalised financial features powered by behavioural AI.

Impact: Strengthened Monzo's fraud prevention and regulatory posture while enabling faster, data-informed innovation in customer-facing financial products.

SEPT 2018 - MAY 2020

AI & Cloud Platform Engineer | The Game Company

TGC is an AI-powered cloud gaming platform. I designed and automated systems that used data and machine learning to enhance gameplay experience, matchmaking accuracy, and player retention.

Key Contributions to The Game Company (TGC):

- Built low-latency AI recommendation systems with TensorFlow and Redis Streams, optimising matchmaking and content delivery globally.
- Architected real-time analytics pipelines for player engagement telemetry, providing game designers with actionable insight.
- Deployed AI-driven platform updates through Kubernetes, ArgoCD, and AWS/GCP pipelines, achieving sub-minute global rollouts.
- Developed reinforcement learning agents to dynamically adjust game difficulty and player experience.
- Worked with data scientists and developers to productionise AI prototypes into scalable, monitored services.
- Improved cross-region performance through CDN and caching strategies integrated with AI load balancing.

Impact: Enhanced player retention and experience by integrating adaptive AI systems into live operations — directly improving engagement metrics and revenue performance

Career Journey into Data Engineering

I began my career with a strong interest in how technology could transform the way organisations deliver services. Early on, I worked in roles that exposed me to cloud, infrastructure, and automation, where I quickly saw how fragmented, manual data processes created bottlenecks for teams and slowed down decision making.

Curiosity led me deeper into data pipelines, cloud platforms, and scripting with Python and SQL, where I discovered how automating ingestion and transformation could unlock faster, more reliable insights. I started contributing to building ETL pipelines, consolidating disparate data sources, and supporting migrations to Azure-based platforms. Seeing how clean, well-structured data improved outcomes for analysts, data scientists, and end users gave me real satisfaction.

Over time, I gained hands-on experience across industries from banking and AI SaaS to defence, gaming, and public services. Each role gave me a broader perspective: how regulated industries value secure, governed data platforms, how high-volume systems demand real-time and reliable pipelines, and how modern organisations depend on data-driven insights to innovate and serve customers better.

Today, as an Azure Data Engineer, I work with a modern data stack Azure Data Factory, Databricks, Synapse, ADLS Gen2, Event Hubs, and Power BI underpinned by DevOps practices such as CI/CD and Infrastructure as Code. I also embrace AI-assisted tools and workflows that have significantly boosted my efficiency: from accelerating SQL and Spark development with AI copilots, to automating testing, monitoring, and documentation.

For me, data engineering is not just about moving and transforming data it's about building secure, scalable, and intelligent platforms that empower organisations to innovate, make informed decisions, and deliver at scale.

May 2017 - Sept 2018

Automation Engineer | ABB (Industrial Automation)

At ABB, I worked on industrial automation projects that combined IoT, AI, and predictive analytics to optimise energy systems and production reliability across manufacturing and utilities clients.

Key Contributions to ABB Digital Sevices:

- Developed IoT data pipelines using Azure IoT Hub and Databricks, connecting thousands of field devices for live telemetry.
- Applied AI-driven predictive maintenance models with TensorFlow and Scikit-learn, reducing unplanned downtime for energy clients.
- Automated training and deployment pipelines through Azure DevOps and Terraform, standardising MLOps across industrial plants.
- Built digital twins of industrial systems, enabling simulation-based diagnostics and performance optimisation.
- Collaborated with operations teams to translate maintenance knowledge into AI rule engines and alerts.
- Implemented edge-to-cloud data security standards, ensuring reliable, compliant transmission of sensitive operational data.

Impact: Improved system reliability and reduced downtime for clients through predictive AI solutions that optimised operations across energy and manufacturing networks.

JLY 2014 - APR 2017

Cloud Engineer | Cargill

One of the world's largest food and agriculture companies, operating across 70+ countries. As a Cloud Engineer at Cargill, I supported the design and automation of AWS based platforms powering food supply, processing, and sustainability initiatives. Focussing on cloud migration, performance optimisation, and security, enabling reliable digital solutions across Cargill's global operations.

Key Contributions to Cargill Supply Chain:

- Supported the migration of supply chain applications to AWS, reducing infrastructure costs by 22% and improving scalability for global logistics and food processing systems.
- Automated infrastructure builds with Terraform cutting provisioning time by 55% and enabling faster delivery of digital solutions to agriculture and sustainability teams.
- Improved system performance and reliability by optimising autoscaling groups, storage tiers, and load balancers, reducing application downtime by 35% and enhancing efficiency in food production and distribution workflows.
- Implemented cloud monitoring and observability using AWS CloudWatch, Azure Monitor, and custom dashboards, reducing mean time to detection (MTTD) and recovery (MTTR) by 30%, giving global operations teams real-time visibility.
- Strengthened cloud security compliance by deploying IAM controls, encryption policies, and automated vulnerability scans, reducing critical security findings by 28% and ensuring alignment with regulatory and corporate standards.

" Azure Data Engineer experienced in building cloud-based data platforms, pipelines, and analytics solutions, with a track record of improving data reliability, security, and accessibility. Open to new opportunities where I can add value, grow my expertise, and help organisations unlock insights and deliver at scale."

Samuel Wasswa Muwonge
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